

Supplementary Material for “PA-LOI: A Proactive Risk Layer for Occluded Pedestrian Emergence”

Hongyu Peng, Shangqi Zhang, Bo Chen, and Xiao Wang

I. LITERATURE POSITIONING

Table S1 complements the main-paper discussion by comparing hidden-risk representation, planner coupling, evaluation scale, and the availability of fixed hazard geometry. The entries are qualitative because the cited systems use different planners and scenario definitions.

TABLE S1
QUALITATIVE POSITIONING OF PA-LOI AMONG REPRESENTATIVE OCCLUSION-AWARE APPROACHES. THE COMPARISON CONCERNS DESIGN CHOICES, NOT HEAD-TO-HEAD METRICS, BECAUSE THE CITED METHODS ARE COUPLED TO DIFFERENT PLANNER STACKS AND SCENARIO DEFINITIONS.

Approach	Hidden-risk representation	Planner coupling	Closed-loop evaluation scale	Fixed-geometry protocol
Set-based reachability [1], [2]	Reachable occupancy behind occluders and beyond sensor range	Verification or constraint layer over a given planner	Formal analysis with case studies	Not released
Dynamic occlusion shadows [3]	Tracked shadow regions with potential occupants	Reactive constraints on speed and clearance	Closed-loop case studies	Not released
POMDP belief planning [4]	Belief over hidden-agent states	Belief-space planner replaces the nominal stack	Simulation case studies	Not released
Occlusion-aware contingency planning [5]	Hypothetical agents with contingency branches	Dedicated safety-critical MPC formulation	Simulation scenario studies	Not released
Visibility maximization [6]	Visibility deficit along candidate paths	Modified objective of the nominal planner	Intersection case studies	Not released
PA-LOI (this work)	Conflict-centered phantom source with safe-speed bound from bounded-rational emergence	Additive velocity-hinge cost in the trajectory optimizer; learned prediction and scenario-tree logic preserved; AEB fallback	4050 closed-loop AV2 replay runs: five systems on 81 scenes at six trigger distances, plus no-ghost, walking-speed, extended-distance, and parameter protocols	JSONL-fixed geometry; archive supplied (records, code, scripts)

II. EXTENDED TRIGGER-DISTANCE RESULTS

Table S2 reports every distance–speed cell from the extended moving-pedestrian protocol summarized in Section V-E of the main paper. The original per-run records are included in the accompanying reproducibility archive.

TABLE S2
MOVING-PEDESTRIAN EMERGENCE AT EXTENDED TRIGGER DISTANCES (27 SCENES, 800 FRAMES). v_{imp}^2 COLUMNS ARE HIT-ONLY MEANS, AS IN THE MAIN-PAPER PEDESTRIAN-SPEED TABLE. BEST LOW-SEVERITY AND IMPACT-ENERGY VALUES PER DISTANCE–SPEED CELL IN BOLD.

Speed (m/s)	Method	$d_{\text{trig}} = 4.5 \text{ m}$			$d_{\text{trig}} = 3.5 \text{ m}$		
		Low-severity	Zero coll.	Mean v_{imp}^2 (m^2/s^2)	Low-severity	Zero coll.	Mean v_{imp}^2 (m^2/s^2)
1.0	PA-LOI + AEB	26/27 (96.3%)	2/27 (7.4%)	1.00	26/27 (96.3%)	0/27 (0.0%)	0.67
	AEB-only	21/27 (77.8%)	1/27 (3.7%)	4.36	20/27 (74.1%)	0/27 (0.0%)	4.83
	MIND	5/27 (18.5%)	4/27 (14.8%)	25.41	6/27 (22.2%)	5/27 (18.5%)	25.01
2.0	PA-LOI + AEB	26/27 (96.3%)	5/27 (18.5%)	2.78	26/27 (96.3%)	0/27 (0.0%)	3.01
	AEB-only	14/27 (51.9%)	1/27 (3.7%)	11.95	13/27 (48.1%)	1/27 (3.7%)	12.81
	MIND	3/27 (11.1%)	0/27 (0.0%)	28.05	3/27 (11.1%)	0/27 (0.0%)	27.25
3.0	PA-LOI + AEB	26/27 (96.3%)	4/27 (14.8%)	3.15	25/27 (92.6%)	0/27 (0.0%)	4.80
	AEB-only	12/27 (44.4%)	1/27 (3.7%)	15.18	10/27 (37.0%)	0/27 (0.0%)	16.41
	MIND	3/27 (11.1%)	0/27 (0.0%)	27.81	4/27 (14.8%)	0/27 (0.0%)	27.40

In one scene AEB-only hits replay background traffic before the trigger in all six cells (counted unsafe); excluding that scene for all stacks changes no rate by more than 3.1 pp and no ordering.

III. RISK-LAYER PARAMETER PERTURBATIONS

Table S3 reports the six one-dimensional perturbations summarized in Section V-F of the main paper. The deployed setting is repeated to make the safety–efficiency changes directly auditable.

TABLE S3
RISK-LAYER PARAMETER PERTURBATION ON THE 27-SCENE SET (GHOST AT 5.5 AND 4.0 M, INSTANT-CENTER, 650 FRAMES, PLUS NO-GHOST EFFICIENCY). THE DEPLOYED ROW USES THE PUBLISHED STRICT-650 TABLES ON THE IDENTICAL CELLS.

Setting	Spawn	Low-severity (all 54 cells)	Low-severity (spawned)	Zero collision (all 54 cells)	Mean v_{imp}^2 (spawned)	No-ghost speed (m/s)	PF/RC
Deployed (0.6/2.0/25)	54/54	92.6%	92.6%	48.1%	2.87	3.47	0/0
$\eta = 0.4$	50/54	87.0%	90.0%	44.4%	2.55	3.29	0/3
$\eta = 0.8$	54/54	87.0%	87.0%	38.9%	3.24	3.57	0/4
$v_{\text{floor}} = 1.0$	51/54	88.9%	90.2%	46.3%	2.72	3.34	0/3
$v_{\text{floor}} = 3.0$	51/54	94.4%	94.1%	50.0%	2.86	3.55	0/1
$w_{\text{max}} = 15$	54/54	81.5%	81.5%	37.0%	3.34	3.54	1/4
$w_{\text{max}} = 35$	51/54	88.9%	90.2%	44.4%	2.44	3.31	0/3

Unspawned cells (ego never reaches the trigger within 650 frames) count as safe in all-cell columns; the spawned-only column removes this credit. PF/RC: planner failures / replay collisions over 27 no-ghost runs; the $w_{\text{max}} = 15$ speed includes one run truncated at frame 630 (3.46 m/s excluding it).

IV. UNFILTERED CANDIDATE-POOL AUDIT

The AV2 search returned 90 new geometric candidates. Two lacked complete three-stack screening records and were not used; 60 of the remaining 88 form new60, while 28 were screened out by the protocol-integrity checks described in Section IV-B. Table S4 removes this clean-scene filter and retains all 88 log-complete candidates. The strict 650-frame evaluable-set rule is applied exactly as in the main paper, background collisions remain unsafe, and non-spawning is exposed by the spawn rate rather than credited as safety. This is intentionally a harsher, confounded pool: the absolute rates fall, but PA-LOI's ordering over AEB-only and MIND remains. The scene-clustered 95% intervals for low severity are [43.0, 61.4], [9.1, 22.8], and [1.0, 9.2], respectively. The archive gives all 90 membership decisions and task-level screening flags; no mutually exclusive exclusion reasons are assigned retrospectively because flags can overlap.

TABLE S4
UNFILTERED NEW90 CANDIDATE-POOL AUDIT. ALL 88 GEOMETRIC CANDIDATES WITH COMPLETE STRICT-650 RECORDS ARE RETAINED, INCLUDING THE 28 SCENES EXCLUDED FROM NEW60. SAFETY IS COMPUTED ON EVALUABLE TASKS; SPAWN RATE EXPOSES NON-ENCOUNTER CONSERVATISM.

Method	Tasks	Spawn rate	Eval. tasks	Low-severity	Zero collision	Background collisions	Mean v_{imp}^2
PA-LOI + AEB	528	73.9%	348	182/348 (52.3%)	47/348 (13.5%)	63	10.15
AEB-only	528	92.4%	471	74/471 (15.7%)	12/471 (2.5%)	15	31.96
MIND	528	98.1%	510	23/510 (4.5%)	0/510 (0.0%)	31	42.06

V. CONFERENCE-TO-JOURNAL EXTENSION

Table S5 makes the extension beyond the accepted ITSC paper explicit [7]. The risk-layer algorithm is not presented as new in this journal submission; the new contribution is the question and evidence enabled by clean81, the five-system comparison, exposure accounting, robustness protocols, statistical analysis, and auditable release.

TABLE S5
ITEMIZED DIFFERENCE BETWEEN THE ITSC CONFERENCE PAPER AND THIS JOURNAL ARTICLE. THE RISK-LAYER ALGORITHM IS EXPLICITLY CARRIED OVER; THE JOURNAL CONTRIBUTION IS THE NEW SCIENTIFIC QUESTION, EVALUATION INSTRUMENT, COMPARATIVE EVIDENCE, AND REPRODUCIBILITY RECORD.

Aspect	ITSC conference paper	Present journal article
Scientific question	Demonstrates whether proactive occlusion-risk shaping can improve on late AEB in representative scenes.	Tests whether the unchanged layer reduces severity with less conservatism than strict occlusion bounds on identical hazards, and quantifies the no-pedestrian cost.
Risk layer	Introduces TTA gating, the crossing-time safe-speed construction, velocity-hinge cost, and AEB fallback.	Preserves those components; adds an implementation-faithful formalization, complete deployed parameters and fallback semantics, and the bounded-rational policy-class interpretation.
Evaluation scale	Two representative AV2-derived occlusion sites with trigger-distance sweeps and an illustrative threat-free trial.	clean81 with 81 fixed scenes, six trigger distances, five stacks, and 4050 closed-loop runs across the main, no-ghost, walking, extended-distance, and parameter protocols.
Comparators	MIND, AEB-only, PA-LOI, and a generic conservative worst-case assumption.	MIND, AEB-only, PA-LOI, and traceable per-step ports of set-based reachability and dynamic-shadow families under shared geometry, physics, and AEB shielding.
Reporting and inference	Collision/impact summaries, one threat-free example, component examples, and module latency.	Low-severity and zero collision separated; background collision, planner infeasibility, impact speed, velocity-squared proxy, spawn-rate conservatism, scene-clustered intervals, paired effect sizes, and exact sign tests.
Robustness and audit	Representative component ablations and qualitative limitations.	Held-out new60 split, thresholds from 2.0–3.5 m/s, moving pedestrians over three triggers, six parameter perturbations, both horizon accountings, and an unfiltered 88-candidate curation audit.
Reproducibility	Scenario-level implementation demonstration.	Canonical JSONL geometry, archived per-run records, method and supervisor code, deterministic table scripts, checksums, and a tested workspace-restoration map.

REFERENCES

- [1] P. F. Orzechowski, A. Meyer, and M. Lauer, “Tackling occlusions and limited sensor range with set-based safety verification,” in *IEEE International Conference on Intelligent Transportation Systems*, 2018, pp. 1729–1736.
- [2] M. Koschi and M. Althoff, “Set-based prediction of traffic participants considering occlusions and traffic rules,” *IEEE Transactions on Intelligent Vehicles*, vol. 6, no. 2, pp. 249–265, 2021.
- [3] Y. Nager, A. Censi, and E. Frazzoli, “What lies in the shadows? safe and computation-aware motion planning for autonomous vehicles using intent-aware dynamic shadow regions,” in *IEEE International Conference on Robotics and Automation*, 2019, pp. 5800–5806.
- [4] C. Zhang, F. Steinhauser, G. Hinz, and A. Knoll, “Improved occlusion scenario coverage with a POMDP-based behavior planner for autonomous urban driving,” in *IEEE International Intelligent Transportation Systems Conference*, 2021, pp. 593–600.
- [5] L. Zheng, R. Yang, M. Zheng, Z. Peng, M. Y. Wang, and J. Ma, “Occlusion-aware contingency safety-critical planning for autonomous driving,” *IEEE Transactions on Cybernetics*, vol. 56, no. 5, pp. 2777–2790, 2026.
- [6] P. Narksri, H. Darweesh, E. Takeuchi, Y. Ninomiya, and K. Takeda, “Occlusion-aware motion planning with visibility maximization via active lateral position adjustment,” *IEEE Access*, vol. 10, pp. 57 759–57 782, 2022.
- [7] H. Peng, S. Zhang, B. Chen, and X. Wang, “PA-LOI: Velocity-aware risk field for proactive occluded pedestrian collision avoidance,” in *IEEE International Conference on Intelligent Transportation Systems*, 2026, accepted, to appear.